Lab 1 – Mastering AI for EEE & Programming

Objective

- To transition from "Answer-Seeking" to "Concept-Mastery" using Generative AI.

Introduction / Briefing

- Instructions: Do not use the AI as a calculator. Your goal is to use the "Three Layers of a Great Prompt" (Persona, Constraint, Task) to deepen your understanding of logic and physical constraints.
- Warning: If the AI gives you the direct answer or final code immediately, you have failed the prompt constraint.

- To do this lab, use Gen AI tool that is available.

Exercise 1: The "Conceptual Bridge" (EEE Focus)

Goal: Understand the *why* behind physical laws rather than just the formula.

- **Scenario:** You are studying **Kirchhoff's Voltage Law (KVL)**. Instead of asking AI to solve a circuit, you want to understand its relationship to the Law of Conservation of Energy.
- **Sample Prompt:** "I am studying KVL. Do not solve a circuit for me. Instead, explain the relationship between KVL and the Law of Conservation of Energy using a water-pressure analogy. Then, ask me a conceptual question to see if I understand how energy is 'lost' in a resistor." ***Try it***.

Step 1: Select a core EEE formula (e.g., KVL, $V=IR$) and write it below.

V = IR

Step 2: Construct a prompt.

- *Requirement:* Ask the AI to explain the physical intuition using an analogy (e.g., water pressure).
- *Constraint:* Explicitly state: **"Do not solve a circuit for me."**
- *Task:* Command the AI to end its response with a conceptual question for *you* to answer.

Your Prompt:

Hello Mr copilot, I am studying V=IR. Could you help me with understanding circuits of V=IR?

Do not solve the circuit for me, instead use analogies like water pressure to illustrate the concept for me. Come up with a conceptual question for me to answer after teaching me.

Your Output: Write down the answer you gave to the AI's conceptual question regarding energy loss or circuit behavior.

AI Qn: Imagine two water pipes connected to the same pump: Pipe A is wide (low resistance). Pipe B is narrow (high resistance). If the pump applies the same pressure to both pipes, which pipe will have more water flowing, and why?

My Ans: Pipe A.

Exercise 2: The "Code Logic Architect" (Programming Focus)

Goal: Improve algorithmic thinking without seeing the final code.

- **Scenario:** You need to write a C++ script to sort a list of sensor data, but you are stuck on the logic.

- **Sample Prompt:**

"I need to write a C++ function to find the average of only the positive numbers in a list. Do not write any code. Provide a **pseudocode outline** or a flow of logic steps. After that, give me a 'buggy' hint of a common mistake students make with loop counters so I can avoid it." ***Try it***.

Step 1: Describe a programming goal (e.g., sorting a list, finding averages).

Sorting a list of numbers in random order to ascending order, C++.

Step 2: Construct a prompt.

- *Constraint:* **Explicitly forbid the AI from providing a code block.**

- *Task:* Ask for a pseudocode outline, a logic flowchart, or a "buggy hint" regarding common mistakes (like loop counters).

Your Prompt:

Hello mr copilot, I am writing a program in C++. It is supposed to sort through a list of randomly arranged numbers and it's supposed to arrange them in ascending order. Please provide me a pseudocode outline or flow of logic, do not write any code for me. After that, give me a "buggy" hint of a common mistake that students make in programming so that I can avoid it.

AI Response Summary: Copy the specific "buggy hint" or logic step the AI warned you about.

A frequent mistake students make is forgetting to update loop boundaries. For example:
- In Bubble Sort, after each pass, the largest number is already in its correct position.
- If you keep looping through the entire list every time without reducing the range, you waste time and sometimes even risk infinite loops if your condition isn't carefully written.

Exercise 3: The "Circuit Troubleshooter" (EEE Focus)

Goal: Develop diagnostic skills by analyzing failure states.

- **Scenario:** You have a theoretical circuit, and the output voltage is zero.
- **Sample Prompt:**

"In a series circuit with a 12V battery and two resistors, the measured voltage across the second resistor is 0V. Give me three different physical 'failure' hypotheses (e.g., short circuit, open circuit). For each hypothesis, explain how I would use a multimeter to verify it." ***Try it***.

Step 1: Define a "broken" state in a system (e.g., "The voltage across the second resistor is 0V").

The voltage across the first resistor is 0.

Step 2: Construct a prompt.

- *Task:* Ask the AI for three physical "failure" hypotheses (e.g., open circuit vs. short circuit).
- *Task:* Ask how to use a multimeter to verify each hypothesis.

Your Prompt:

Hello mr copilot, right now, I have a series circuit consisting of a 12V battery and 2 1k ohm resistors. Measuring the voltage across the first resistor, it read a 0. Could you give me three physical "failure" hypotheses in order for me to troubleshoot it ? How do I verify each of the hypotheses with a multimeter ?

Diagnostic Plan: List one verification method (e.g., multimeter setting and placement) suggested by the AI.

Battery or Wiring Issue
- Hypothesis: The battery isn't delivering voltage to the resistor (dead battery, reversed polarity, or broken wire before the resistor).
- Verification:
- Measure voltage directly across the battery terminals—it should read ~12V.
- Then measure voltage at the resistor's input lead relative to ground. If it's 0V, the battery isn't reaching the resistor.
- Trace continuity from battery positive to the resistor lead with the multimeter

Exercise 4: The "Edge-Case Stress Test" (Programming Focus)

Goal: Learn to anticipate errors and improve code robustness.

- **Scenario:** You've written a basic calculator program in C++.
- **Sample Prompt:**

"I have a function that takes two integers and divides them. I want to improve my 'defensive programming' skills. List 5 'edge cases' or 'malicious inputs' that could crash this function or produce logical errors. Explain why each one is a threat." ***Try it.***

Step 1: Paste a brief function description or logic idea below.

Function where it takes 3 integers and adds all of them up.

Step 2: Construct a prompt.

- *Persona:* Ask the AI to act as a **Quality Assurance (QA) Engineer**.
- *Task:* Request a list of 5 "malicious inputs" or edge cases (e.g., Null values, Overflow) that could crash the system.

Your Prompt:

Copilot, you are now a Quality Assistance Engineer of C++ Code. Help me troubleshoot my code. I have made a calculator in C++ where it takes 3 integers and adds all of them up. Help me list 5 'edge cases' or 'malicious inputs' that could crash this function or produce logical errors.

Risk Analysis: List two specific edge cases the AI identified that you had not previously considered.

1. Uninitialized Variables
2. Integer Overflow

Exercise 5: The "Reverse Instructor" (Comprehensive)

Goal: Test your own knowledge through peer-review.

- **Scenario:** You want to check if your summary of **Thevenin's Theorem** is correct.
- **Sample Prompt:**

"I am going to explain Thevenin's Theorem to you as if I am the teacher. I want you to act as a student who is confused. Critically analyze my explanation for any technical inaccuracies or 'missing links' in logic. Tell me where my explanation is weak." ***Try it***.

Step 1: Choose a complex topic (e.g., Understand C++ pointers) and write a brief explanation of it below in your own words.

C++ pointers, a pointer is a variable that stores the address of another variable.

Step 2: Construct a prompt.

- *Persona:* Tell the AI to act as a **student who is confused**.
- *Task:* Ask the AI to critically analyze your explanation for "missing links" or weak logic.

Your Prompt:

Copilot, pretend you're a confused student in understanding C++ pointers. I am going to act as a teacher, describing C++ pointers, critically analyse my explanation for weaklinks or weak logic. C++ pointers, a pointer is a variable that stores the address of another variable.

Critique & Refinement: What specific weakness did the AI find in your explanation? How did you correct it?

Too minimal. - The definition is technically correct but very bare. It doesn't explain why pointers matter (e.g., accessing memory directly, passing large data efficiently, dynamic allocation).

Reflection Question for Students:
How did your understanding of the topic change when you were forbidden from asking the AI for the direct answer?

I had to grasp the topic first before the AI spoonfed me with answers.

The AI Study Mentor: Power-Prompt Cheat Sheet

1. The "Physical Intuition" Template (EEE)
Use when: You know the formula but don't understand the physical behavior.
"Explain the concept of **[Insert Topic, e.g., Impedance or Capacitance]** without using any math first. Use a real-world physical analogy (like water, traffic, or heat). Once I confirm I understand the analogy, show me how the standard formula **[Insert Formula]** represents that physical behavior."

2. The "Logic First" Template (Programming)
Use when: You have a coding task but don't want to skip the thinking process.
"I am working on a **[Language, e.g., Python/C++]** program to **[Insert Task]**. Before I write any code, act as a lead developer. Provide a step-by-step logic flowchart in text form. Identify where I should use **[Insert Concept, e.g., nested loops or recursion]** and explain why that is the most efficient choice."

5. The "Socratic Exam Prep" (Comprehensive)
Use when: You think you know a topic and want to be tested.
"I am studying **[Insert Module Topic]**. Act as a professor. Ask me one challenging 'high-order' conceptual question that requires me to apply **[Concept A]** to **[Concept B]**. Once I answer, critique my technical accuracy and tell me if I missed any critical engineering constraints."

Why this works:

- **Active Recall:** It forces you to retrieve information from your own brain.
- **Deep Encoding:** It connects abstract programming concepts to logical structures.
- **Verification:** It mimics the "Human in the Loop" approach used in professional AI development.

~ The END ~